

ORIGINAL RESEARCH

Researchers' perspectives of accessing healthcare systems data for use in randomised controlled trials

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Abstract

Objectives: Identify the key challenges and opportunities researchers face when accessing healthcare systems data (HSD) for randomised controlled trials (RCTs) and develop a consolidated set of recommended improvements.

Methods: Multidisciplinary researchers with expertise in RCTs were invited to participate in online focus groups and individual interviews to share their experiences of accessing HSD. Data were analyzed using directed content analysis guided by themes identified in prior research.

Results: Three distinct stages of the HSD access process were identified, covering preparation, submission of a data access application to a provider, and working with the received data, each presenting specific challenges and opportunities for improvement. In the preapplication stage, participants highlighted limited awareness and experience with HSD and emphasized the need for clearer guidance. During the data access application stage, researchers described difficulties related to the mismatch between application requirements and the needs of RCTs, as well as delays in progression. In the postapplication stage, challenges in receiving and working with the data were reported, alongside recognition of the tangible benefits that HSD can provide for trial delivery and quality.

Conclusion: Identifying challenges across the HSD access process can help trial teams anticipate potential obstacles and plan more effectively. Given that the current application process does not fully meet the needs of RCTs, the recommendations derived from this work highlight key areas for improvement that providers could implement to reduce barriers and better support the use of HSD in trials. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Routinely collected data; Clinical trials; HSD access applications; Healthcare systems data; Real-world data; Electronic health records; Randomised controlled trials; Randomized controlled trials

1. Introduction

Healthcare systems data (HSD) refers to medical information collected without having a specific research question formulated in advance. Such data can be gathered from different sources, including centralized data providers, like National Health Services (NHS) England, disease-specific patient registries and individual health care sites,

like specific hospitals. Access to HSD sources has the potential to increase efficiency of randomised controlled trials (RCTs), the gold standard for evaluating health care interventions [1], by providing access to a large amount of data, while decreasing participant and staff burden [2].

The use of HSD in research is increasing [3], and its benefits and limitations in RCTs are being explored worldwide [2,4–6]. In 2022, 62% of the on-going National Institute for Health and Care Research (NIHR)-funded trials were planning to use HSD, showing an increase of 15% compared to 2019 [7].

In addition, several clinical trials funding bodies, including the NIHR, have been requesting, within their applications, considerations regarding the possibility of using HSD and launching specific funding calls for data enabled trials – a feature of which is used for HSD [8].

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What is new?

Key findings

- The current healthcare systems data (HSD) application process is not fit for the purpose of randomised controlled trials (RCTs), with issues including unsuitable requirements, delays, and limited mutual understanding between trial teams and providers.

What this adds to what is known?

- This study provides recommendations for trialists applying for HSD for use in RCTs, aiming to ease the current process by highlighting common barriers and facilitators.

What is the implication and what should change now?

- The development of an updated HSD applications is needed to account for RCT requirements.
- Improved communication and transparency between trial teams and HSD providers could reduce delays and inefficiencies.

In 2022, a Delphi study was conducted to identify and prioritize the most important questions to be answered about use of HSD in RCTs [9]. The prioritized questions focus on the following topics: data access and receipt, data collection method, outcome selection, communication with participants, regulatory approvals, data quality, and data

analysis. Based on this study, several streams of action have been implemented, with the areas of data collection and communication having been addressed [10,11] and ongoing work to assess the data utility of HSD, by comparing it to bespoke trial data [12]. The current work expands the areas of data access and regulatory approvals.

A study conducted in 2023 looking at determining when the use of HSD is appropriate established a list of key issues to consider at the design stage of an RCT [10]. The feasibility of using HSD in RCTs was highlighted among the key topics. This item included several issues to be considered regarding the process of accessing data: seeking to include in the trial team specialists with experience of completing data access forms; establishing whether HSD includes what the trial needs; gathering data quality insurance from the provider; discussing resolution of data errors and discrepancies with the data provider; and considering the timelines for receiving the data from the point of approval. Figure presents the process of accessing HSD for use in an RCT.

Within the UK, access to HSD is not standardized across systems, with each HSD provider requiring a detailed application in order to release data. The application process differs depending on the data provider but usually requires describing the purpose and proposed use of data, discussing the study details with a case officer and establishing a data sharing agreement (DSA).

Several recent publications looking at the use of HSD in RCTs, have highlighted the process of accessing data as a topic of interest and one of the challenges that may discourage HSD from being used more routinely [9,10,12]. Previous research has described some of the issues associated with accessing data, when HSD is to be acquired alongside other forms of routine data, by examining the progress of three RCTs [13]. The current work aims to

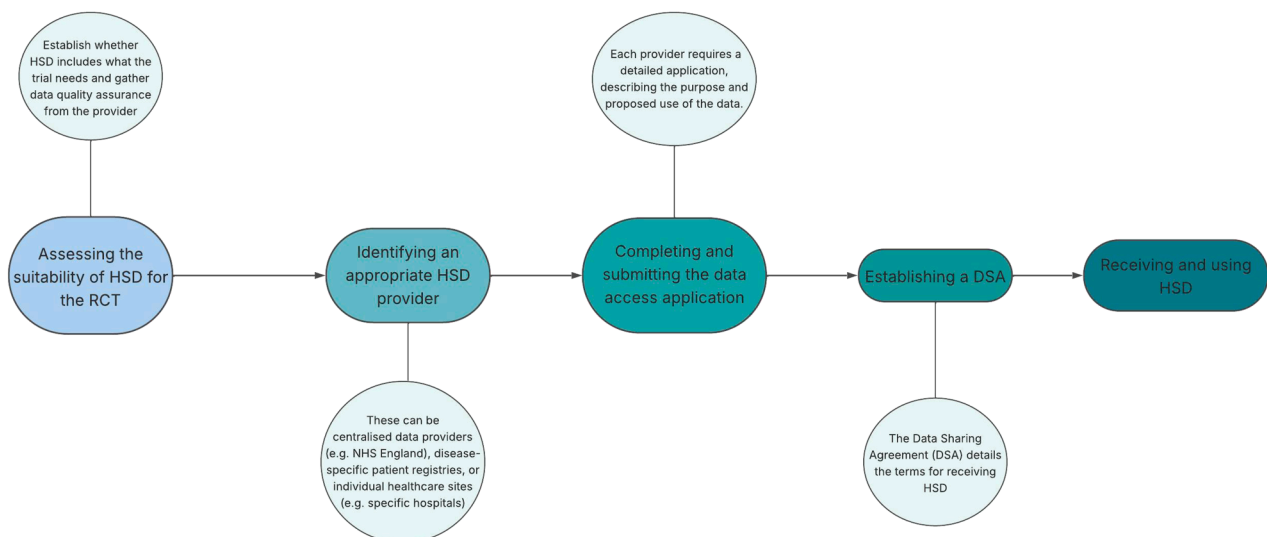


Figure. Overview of the HSD access process for use in RCTs.

uncover and address the current challenges encountered during the application process, by appealing to researchers with expertise in different areas of RCTs and bringing together a list of recommended improvements.

2. Methods

2.1. Recruitment of participants

Purposive sampling was used in order to ensure perspectives from participants working in different roles within a trial were represented within the sample. This form of non-probability sampling, where the researchers choose the population to participate, means that the participants are selected to have a wide range of involvement in the process, in order to present the challenges encountered from all aspects of accessing HSD. All participants were given a participant information sheet and provided their consent prior to their interview or focus group.

2.2. Data collection

A mix of focus groups and interviews were conducted. The use of qualitative methods allows for an enhanced understanding of the current issues. The focus groups allowed multiple perspectives to be captured, while the open conversation extended the discussion area, therefore highlighting different topics of interest. The groups were determined based on the researcher's availability, while ensuring a spread of experience where possible. The participants, unable to attend the focus groups, were invited to participate in individual interviews. The focus groups and interviews were conducted online, audio-video recorded, and transcribed verbatim. Focus groups lasted between 60 and 90 minutes and individual interviews between 30 and 60 minutes. The same topic guide was used across both formats.

The topic guide used during the discussion was developed focusing on the four questions to be answered within the study:

1. What are the expected benefits from using HSD?
2. What challenges have been encountered within the accessing process?
3. What current practices are helping the process?
4. What improvement would impact and ease the process?

These questions were formulated to offer a rounded view of the application process, guiding the discussion through the challenges and opportunities encountered.

2.3. Data analysis

A directed content analysis was undertaken. This consists of three main phases, used to analyze qualitative data

by using core themes, identified from previous work, as guidelines, and allowing new codes to emerge from the data, where the content did not fit predefined categories [14,15]. Author's contributions in data analysis are identified with initials.

Phase one encompasses three sequential steps, which prepare the data for analysis. Step one is to develop a guide coding framework based on pre-existing themes. The themes used were formulated to follow the aims of the study and informed by previous literature. They are presented within the coding guide attached, available in [Appendix A](#) and describe the areas of interest (A-MT, CG). Step two is to determine the materials to be analyzed, which in this case were focus group and interview transcripts. This involved purposefully sampling the population and deciding on the data collection method (A-MT, KG, and CG). Step three is to get familiarized with the data by going through the materials several times (eg, reading transcripts) and identifying text of interest. (A-MT)

Phase two describes the in-depth stages of data analysis. Step four involves coding and organizing the data into the predetermined themes from step one (A-MT, CXX). Step five identifies the connections between the themes and interprets them, by building a logical chain of evidence. This step involves generating new codes (A-MT). Step six describes the findings by explaining the analytical process. This is supported with quotes and tables from the data. (A-MT, KG, CG)

Phase three acts as a guide to reporting the results. Step seven creates an outline for the conclusions. The results present the initial themes alongside explanations of any subsequent alterations and newly generated codes (A-MT, KG, and CG). Step eight is to write the final description of the research history and the findings (A-MT, KG, and CG).

All transcripts were pseudo-anonymized before being imported into the data management system, NVivo, which was used to help organize and code the data. An independent researcher with experience in working with qualitative data reviewed the coding for 10% of the transcripts.

The comments made during the focus groups and interviews have not been attributed to the participants, therefore no purposefully identifiable information is disclosed.

The SRQR (standards for reporting qualitative research) checklist was completed to present the reporting of qualitative research and is included in the [Supplementary materials](#).

2.4. Ethical approval

This study was approved by the Institute of Population Health Research Ethics Committee at the University of Liverpool (Reference: 14,940). Informed written consent was obtained from all participants prior to the interviews or focus groups.

3. Results

3.1. Participant characteristics

In total, 20 researchers (termed participants from here onwards) from 15 organizations within the UK attended either a focus group ($n = 3$, ranging from 3 to 5 participants per group) or an interview ($n = 9$). The participants held a wide range of roles, within RCTs, including statistician, trial methodologist, and trial manager, or with a data provider, and had varied experience working with HSD. The participants' demographics are shown in Table 1.

3.2. Findings

The analysis of the discussions highlighted the existence of three distinct stages in the HSD access process that have specific considerations for trialists, containing particular challenges and opportunities. These stages are: (1) preapplication, showcasing the particularities of the decision process required before access to data was requested; (2) application, describing the issues that arose while trying to set in place an access agreement; and (3) postapplication, explaining the process of receiving the data as well as the experience of using it. These stages and associated themes are presented below.

3.2.1. Preapplication

The participants reported several challenges and opportunities around the decision to use HSD for RCTs. These were linked to the reasoning behind choosing HSD, the issues encountered at this stage and the improvements considered to ease the process.

Table 1. Participants' demographics

Characteristic	Number
Role ^a	
Statistician	9 (45%)
Trial methodologist	4 (20%)
Trial manager	3 (15%)
Data provider	3 (15%)
Clinician	2 (10%)
Health economist	1 (5%)
CTU director	1 (5%)
Data manager	1 (5%)
Experience in RCTs	
Less than 10 y	5 (25%)
More than 10 y	15 (75%)
Number of trials worked on using HSD	
Less than 5	11 (55%)
6–10	5 (25%)
More than 10	4 (20%)

^a Some researchers reported holding multiple roles.

3.2.1.1. Reasons for choosing to use HSD. There were two main reasons that influenced the decision to use HSD. Firstly, participants noted a perceived expectation from funders to consider whether HSD can be used. Beyond the funding programs, specifically asking for RCTs using HSD that are currently available, particular funding bodies are said to require consideration of using HSD and expect reasoning to be presented where it cannot be used in all applications.

“So sometimes it's the funder. Like, part of their funding brief, will expect or ask for a consideration of if routine data can be used, and especially for sort of more longer term follow up or consent for longer term.” (P18, Interview)

Secondly, the potential benefits of using HSD were discussed as a deciding factor. They relate to both the available dataset and to reducing some of the burden that RCTs entail on participants and sites. From a recruitment perspective, there was a belief that using HSD in an RCT may allow access to a wider, more inclusive, pool of participants, enabling larger trials and the ability to better understand the population.

“I mean for us, when we were doing a large scale trial, it definitely helped us to access the data to, to help with the recruitment to assess eligibility.” (P8, Focus Group)

Participants also discussed that a decrease in the burden imposed on trial participants is a potential consequence of using HSD for outcomes, as it minimizes the number of trial specific visits. This reduction was also considered a way to decrease the burden for clinical staff that would otherwise have to collect each outcome at various time-points from the participants, therefore also minimizing data collection duplication.

There was a belief among interviewees that the data to be collected from HSD can offer access to a wide variety of outcomes. As the data is generated independently of the trial, it can minimize the missingness of the trial dataset while improving participant follow-up,

“For our patient group, because we are treating mostly frail, older individuals, we do tend to lose them in terms of follow up. So you can't even figure out whether they have died or not. If you're looking for five and 10 year data. So it's impossible to deliver a vascular surgery trial without getting some form of access to existing data about death, amputations, etcetera.” (P13, Focus Group)

In addition, the assumed accuracy of the data, as well as an assumed cost reduction related to using HSD were debated during the discussions. While the idea of using HSD in order to access more accurate data was brought up by several participants, whether HSD is of sufficient quality to support RCTs was pinpointed as a topic of

interest reported by a few participants, as not all data are equally validated. The necessity of evaluating the difference in the cost of a trial using HSD against one using bespoke data was also noted, as participants wondered about the supposed cost reduction.

“Yeah, I was just going to say that the costs associated with paying [Data provider 2] to do their data whatever they do, was meant to be cheaper than setting up a bespoke system, but I mean, I haven’t done the maths, but it would have been a lot more efficient, quick and just to do it ourselves I think.” (P2, Focus Group)

3.2.1.2. Issues encountered. There were several issues reported in relation to the pre-application stage, referring mostly to lack of knowledge and experience with HSD. Firstly, there were challenges encountered due to insufficient information about the process of accessing data. Although there is guidance available, its lack of visibility and its interpretability was noted.

“Quite often teams, if they’re doing the application themselves, they don’t know and they come to me and ask me what they need to be putting in. So often it’s a lack of knowledge about what is required.

And lack of knowledge of where to look because there is guidance available in different places. So [Data provider 1] have their list of standards, but often people don’t realise it’s there and they don’t really understand what they mean as well. Sometimes interpretation is difficult.” (P14, Interview)

Secondly, a lack of information regarding the availability and quality of the data was mentioned as being problematic. Many of the participants described issues related to requesting the wrong codes from HSD or requiring assistance in choosing the right ones, as the data dictionaries might not provide all the information required to make an informed decision of selecting the data of interest.

Lastly, the lack of centralized providers, which may enable access to data from multiple UK nations and various datasets, was highlighted as a deterrent from using HSD. It was noted that RCTs that require obtaining data from several HSD sources, need to go through the application process multiple times, as each HSD source has specific requirements, which need to be considered from the design stage of an RCT. In addition, this deters trial teams from running studies across the devolved nations, as each system would require an additional application.

“So one hurdle firstly is that that is only England data, so clinical trial recruiting patients from Wales or Scotland or Northern Ireland. You know they would be missing data unless we approached the equivalent organisations in those countries as well.

So immediately that doubles or even quadruples the workload because we have to approach each and everyone.” (P17, Interview)

3.2.1.3. Suggested improvements. Several improvements to ease the challenges encountered in the preapplication phase were suggested during the discussion. These map onto the issues described previously (“b. Issues encountered”), aiming to help the process.

The availability of more detailed data dictionaries was considered to potentially ease the data selection process significantly. This would need to include significant details about each data field, as well as examples of the data available. Alternatively, the possibility of having dummy datasets was noted. This would show both the quality of HSD and the data within.

“And then the other thing that some providers do is having like a synthetic dummy data set.

Which is really helpful as well so that you can actually see how the data is formatted.” (P9, Focus Group)

The concept of having unified access to HSD either at national level or UK wide was noted by all the participants as an opportunity for improvement. This could allow for a single data access application with a single set of requirements. Participants believed this could minimize the current deterrent caused by the need for individual processes for each data source. In addition, this unification could allow for a more complete view of the participants’ records, documenting their journey within the medical system.

“It’s probably nobody’s priority to join that up, but if it were, it would be really helpful. I mean, the dream would be if your primary care records and your secondary care records were all integrated.

I mean, even as a patient, there are times when you sort of think: Wouldn’t it be nice if these systems spoke to each other?

So, you know, certainly as a researcher, wouldn’t it be nice if the system spoke to each other? You know, there are ways and means of linking things up, but it is tricky and it’s not, just because you solve it for one study doesn’t mean you’ve solved it for the next one, and the next one.” (P4, Interview)

Lastly, reaching out to other peers with experience in accessing HSD or discussing with groups with experience in data access from HSD was reported to significantly help ease the process. It was also recommended to consider adding someone with HSD access experience on the team when designing the trial, as they could guide the trial team through the process.

“I think you get to speak to people who have gone through this before and you learn these challenges before facing them yourself. You know, trying to learn from other people’s mistakes and and, you know, I think that has to be the best way.

Make use of you know the end of the phone and people are willing to to assist and help.

Difficult to struggle by yourself, you know. Pick up the phone and ask [Data provider 1] about this, that and the other. And and they are very helpful.” (P17, Interview)

3.2.2. Application

During the discussion, the participants highlighted that many issues were encountered regarding the application process. Possible improvements, both at HSD source level and actionable by participants, were noted as having the potential to ease the process.

3.2.2.1. Issues encountered. The application process to access HSD was considered to be long and difficult. The process was described as being iterative, with the application being sent back and forth many times between the trial team and the data provider. Issues arising from the wording used when requesting participant consent and lack of confidence in the trial team interpretation of the HSD provider’s requirements have been mentioned among the reasons for the prolonged application.

“I think sometimes you have people who don’t entirely understand the nature of the work. And clinical trials are quite specific.

[...]

If your patients have consented, then it should be fine to be doing the work, but in practise there’s quite a lot of, sort of internal knot, like your data provider will want to just, you have to justify your consent. You have to justify everything and it’s it seems like often they’re trying to kind of double guess a lot of what you’re what you’re doing and what you’ve already done.” (P16, Interview)

Timelines were considered to be extensive. Changes in the regulatory environment (for example, when general data protection regulation (GDPR) came into effect) or within the data provider’s requirements resulted in a need for reapplication. Delays in data access were noted to generate a range of subsequent issues including staff turnover, both at trial team and data provider level, increased costs, and a need for alternative plans for data collection.

“You will have applications that were submitted under one system and they sort of say well too, too much time has lapsed, you’ll need to resubmit it now using a different set of forms or I’m really sorry your case manager has left. So we’re giving you this person. Could you please just explain the thing all over to them?

[...]

And actually we have one where we’ve been getting regular sort of annual updates and some process has changed and now they said, I’m sorry to get the final extract you’re going to have to reapply all over again.” (P4, Interview)

The participants mentioned that when HSD was requested from multiple providers regardless of the country, the number of applications required increased. While there are similarities between the applications, there are differences. It was noted that the wording required and the amount of information requested to answer each question differs for each provider.

“They, I mean, I wouldn’t say any of them are necessarily better than the other. The principles of the application form across all of them are the same. But the way in which they want the information or the word count that you have in particular sections, it inevitably differs from one to another. The speed with which they review them and the depth with which they review them and provide comment, varies hugely amongst all of the different providers of data.

Yeah. So, they, they’ve all, they’ve all got subtle differences, but the kind of overarching principles of all of them is the same.” (P5, Interview)

Participants highlighted that current HSD applications are not tailored for trials, and the rigidity of the systems does not allow for deviations and considerations to be in place. Some participants noted that, as clinical trials are already regulated by the Health Research Authority in the UK, the current HSD application is too extensive and not suitable, duplicating efforts to ensure already monitored good practice.

“But when I’ve even tried to apply to [Data provider 2] to obtain data for a study that they’re being paid to work on, that is being randomised through their systems, their request, their forms for applying for their data are not set up for trials, and so it was in a massive headache to convince their team that they should provide us with the data that they’ve already said they’re going to provide us with, you know, it

was just a bureaucratic waste of time, so.” (P18, Interview)

An additional issue that was reported related to consideration regarding the storage, archiving, and sharing of data. Although RCTs require trial teams to retain their data for a long period of time, of up to 25 years, HSD providers are often said to ask for the data to be destroyed within a shorter window, for example, 6 months after the initial download. The participants reported that some HSD providers are moving toward using trusted research environments (TREs), where the trial team can access the data without being able to store a copy in their trial systems. Onward data sharing (ie, for secondary analysis) was also mentioned as a consideration of interest that needs to be discussed with the provider in advance, as it depends on specific characteristics of the trial (such as the planned collaborations with external institutions) and the data to be accessed.

“Some of the trials we’ve we’ve found difficult because you can’t always share data. That’s maybe slightly to one side of this, but if you’re working with an external funder or manufacturer and you have a legal and contractual obligation to provide them with data, then using linkage data for that becomes very complicated and I’ve got an application at the moment that I’m going to have to start, which will be specifically to onward share with the the manufacturer and the funder, and that will probably take the best part of a year to get in place.” (P16, Interview)

3.2.2.2. Suggested improvements to be implemented at data provider level. In order to address the issues previously described at the application stage, the participants suggested several improvements. Realistic timelines were noted as a development of interest, as unexpected delays in receiving the data can impact the trial progress. The trial team’s lack of knowledge regarding the steps their application is going through and the provider’s lack of knowledge regarding trial regulations were highlighted as important obstacles that can be alleviated by building an understanding of the processes, through enabling communication. Other suggestions for minimising the bureaucracy included application unification between providers, having more examples of successful applications for trials, and the possibility of running trials within centralized health care systems, that is, NHS England.

“So I mean, I completely agree that the application processes have been like the biggest challenges. They’re so different. But well, I’ve only got experience between England and Scotland, but they’re so different between the applications, all asking completely different things. And if there’s some way where we could link, you get an overarching for the four and then a slightly shortened version for each of the four nations, then would be fantastic,

but four separate applications all asking different questions is very difficult to manage and navigate.” (P10, Focus Group)

The need to bridge the language gap between data providers, trial teams, and clinicians was suggested several times. It was highlighted that several terms are used by each group to describe different concepts. This discrepancy in terminology was noted to inhibit communication, and consensus of wording was suggested as a potential resolution.

“You know, so we used to have, say, oh, we’re screening the patient list to mean reviewing [....], so we send a list of, they’re basically local identifiers, so that the practise can sit down looking at their electronic CRF [case report form] and see the [Data provider 7] numbers for the patients they’ve got and then they can go on to their [Data provider 7] system and and review them to see whether or not they’re suitable. Review is the most accurate word for that. You can’t call it screening because they haven’t actually consented yet.

But when I joined [redacted], everybody was calling it screening and yeah, I did go blue in the face quite a lot saying, look, it’s really, you’ve got to stop saying screening because if the research networks here you say screening pre consent, they are going to have a fit.

So yeah.

I think some, yeah, common language is definitely an issue.” (P20, Interview)

3.2.2.3. Suggested improvements at researcher level. While many of the improvements suggested were dependent on changes to be implemented by the HSD provider or systems, several participants provided advice for trial teams planning to access HSD in the future. This related to planning in advance and designing their RCT in accordance with HSD provider regulations. Several existing resources were mentioned in the discussions, including published journal articles and contact details for support groups.

“I think and also I’d say plan.

Don’t think that it’s going to be a quick process because it won’t, so trying trying to speak to [Data provider 1] well ahead of it, probably in the grant application stage. Just so they know it’s coming. If it’s funded, then and at least you can start contracts and going through that process because there’s different committees, it’ll go in front of to see whether. Because obviously they’ve got to make sure that they’re that we’re not collecting any data that is unnecessary and that we’re following the principles

of, yeah, data protection as well, so.” (P19, Interview)

3.2.3. Postapplication

The postapplication stage gathers the challenges and opportunities encountered after the application process is completed and a data sharing agreement is established. The following section describes the issues, alongside considerations for potential improvements.

3.2.3.1. Issues encountered. There were several issues noted relating to the process of receiving the data and its format.

Firstly, the limited notice when data are available to be downloaded was mentioned by several participants. An email is said to be sent manually to a single person within the trial team informing them of the availability of the dataset. The limited time allowed for data download can cause issues in case the contacted trial team member is unavailable, and the restrictiveness of the system limits the possible mitigations.

“You know, as soon as someone puts the data across onto [secure file transfer system], you know, they’re secure, sort of transfer of file file transfer software they you know, there should be an automated, automated kind of e-mail to say you’ve received data, but no, they do it manually.

So the data could be sitting there for weeks and you don’t know.” (P14, Interview)

Alternatively, some providers are said to now offer their data through TREs, which allows for data analysis to happen online. Only results are to be downloaded from the system, but several participants noted that there are various restrictions on the type and amount of information to be exported. For example, one researcher mentioned that when the number of events is smaller than ten, the particular result cannot be exported from the TRE and reported.

Secondly, further delays were encountered due to data release lags and linkage of datasets. Where multiple datasets were requested from the same HSD provider and if the linkage was to be performed by the provider, some participants reported experiencing delays. In addition, it was noted that if an HSD source requires the use of their TRE, the trial team might be asked to upload datasets received from the other sources, including other HSD providers, onto the TRE for linkage, which may cause additional delays.

“I would say there are always subtleties involved, but one issue I think we have had, and again, this might be something specific to the circumstances we’re in, is often when I’m trying to use data across [UK country 1] and [UK country 2] because the [UK country 2] data has to stay in the [Data provider 5’s TRE], we

have to import it from [Data provider 1] into [Data provider 5’s] TRE.” (P1, Focus Group)

Lastly, when discussing the quality and format of the data received, some of the participants reported having experienced issues related to missing data and the amount of formatting required to produce an analysable dataset. A few participants noted that the quality and stability of data recorded for research purposes may differ from the one to be expected within HSD. In addition, the participants highlighted that they had experience of receiving incorrect or not fully anonymized data.

“So yeah, so I mean in the past or actually [Data provider 1] with their [Database 1] data still send data out rather than using it in a secure data environment. And I’ve lost count of the number of times that they have sent us the wrong data.

So, so that does tend to be a particular [Data provider 1] problem.

So whether it’s, you know, wrong dates or they pulled the wrong information based on the codes we sent them or things like that.

Yeah, it’s probably been the biggest challenge.” (P5, Interview)

3.2.3.2. Improvements suggested. In order to help alleviate the issues encountered in the postapplication stage, participants suggested some improvements. The existence of automated data release systems was suggested to ease possible delays caused by manual data extraction. It was noted that several databases already have this in place, and the data are released quickly.

To account for the issues related to the data quality and completeness, the availability of information regarding the completeness of the data was discussed. Additional investments in the quality were also mentioned, including the possibility of having data validation per field at point of collection.

“You know, they’re investing and they’re they are encouraging researchers to use [Database 3] in the name of efficiency and et cetera. But they can only do that if they actually invest. If the government invests in making sure that the data are reliable so you can’t have best of both worlds. And so it you know, it just I mean it’s never going to happen because the [Data provider 1] databases and things are so huge, but there needs to be validation on the fields. There needs to be, you know, a reminder that if this field hasn’t been completed, does that mean the event hasn’t occurred, you know and the amount of effort that goes into that on a trial level shows why that’s never going to happen in practise, but if they

want us to get to the stage where we can basically run trials through [Database 3] than they need to invest more in the reliability of the data.” (P2, Focus Group)

It was also suggested that clean datasets be made available. In some cases, HSD dataset has been already cleaned and formatted for analysis by a team, either at trial or provider level. It was considered useful if the provider would allow for this dataset to be shared in the future, as it would minimize duplication of effort regarding data cleaning.

3.2.4. Good experiences

While the process of accessing HSD for an RCT can be challenging, participants noted that getting in contact with data providers early on improved their experience. Invaluable help offered by the data provider’s case officers allowed many trial teams to successfully access HSD, by providing guidance during the application process, helping with choosing the data codes, improving understanding of the requirement, and supporting the trial team through the application.

“When we got assigned a case officer that was really positive, they were really, really helpful in terms of helping us to write the data sharing agreement, what we needed to put in there, kind of sent them a draft.

They also talked us through next steps for when we wanted to do our follow up data extract. So once you get through to a case officer, they’re really, really efficient in coming back to you really prompt with their responses and really supportive as well.

When they took it to panel and there were a few questions that came from that, and actually they answered them in the panel meeting on our behalf, just because we’d already built up such a good relationship and they’d kind of gone through what we wanted.” (P12, Focus Group)

The participants noted some benefits of using HSD, which included the richness of the dataset, as it offers a volume of information that could not be gathered through bespoke methods of data collection.

I think for me is, you know eventually when you do get the data it is great to get the data, and the data generally really adds depth to your study. So I think once you actually can just pass over the awfulness of being able to get it once you get it, it’s great and it’s quite when you finally get it, they tend to be quite responsive and they send you things quite quickly eventually. (P3, Focus Group)

Building good relationships with the data provider’s teams was also a reported benefit. This was noted to help with future application, as an initial point of contact in

planning future studies, at the preapplication stage. Some participants also mentioned that postapplication, after having the agreements set, the data turn-around has been very good.

When it’s worked, when it’s been done quickly and getting it sooner than you expect is amazing. That has happened a few times. (P14, Interview)

4. Discussion

4.1. Summary of main findings

The current research explores barriers and opportunities of accessing HSD from different perspectives. The themes of interest related to the factors that determined the use of HSD at key stages of the application process are the challenges in accessing and receiving the data, positive experiences, and possible improvements.

Key focal points of the discussion included the unsuitability of the applications considering the particularities of RCTs, the issues caused by the differences in application between providers and the limited shared knowledge regarding their respective approaches between trial teams and HSD providers.

The current systems for requesting HSD are very rigid and do not accommodate the needs of RCTs, resulting in enhanced challenges for trial teams. A lack of confidence in the trial team’s interpretation of requirements from HSD providers was reported by several participants, as they consider that there is a lack of understanding of the rules and regulations RCTs are subjected to. HSD applications were perceived as unjustifiably extensive for the RCT context by most participants. While there is a lack of literature on the terminology difference between health care providers and trial teams, similar issues have been explored in other areas of clinical trials, showcasing issues that can arise, for example, having variable usual care being inconsistently delivered to patients and suggesting standardized terminology [16,17].

The lack of data provider application unification was discussed by most participants. The repetitiveness of the questions in slightly different formats for each HSD provider was considered an issue, as participants noted that while the information to be requested is the same, the amount of work required to fill in each application is extended due to minor differences in wording or formatting (eg, different word limits or different key word requirements). This issue has been previously identified and reported for RCTs requesting HSD alongside other routine data; the advice was to allow appropriate timelines and to discuss with peers [13]. This recommendation aligns with our findings, with participants suggesting that increasing time requirement to allow for this and reaching out to others with experience in completing the application can ease the process. Achieving broader unification at national

or UK-wide level, however, would require engagement from government or national health policy bodies.

When discussing the data received from HSD, most participants agreed that the quality is often sufficient and offers the RCT a depth of understanding and several issues were identified. Challenges related to timeliness were reported by all participants. The delays experienced can stop the trial teams from effective monitoring of their participants safety and data integrity. A review discussing the barriers of commercial clinical trials has previously identified the delays in accessing HSD to be a major contributor to trial inefficiency [18].

Several participants noted having received the wrong data. The examples given varied from receiving data for the wrong patients to receiving information from outside the specified date of interest. While on-going efforts are assessing the suitability of replacing bespoke data with HSD for RCTs [12], there is a lack of literature around the confidentiality issues caused by receiving incorrect data. In line with our findings, previous studies noting this issue recommend additional checks be set in place to ensure the validity of the data [19,20]. The use of TREs was discussed as a resolution for trial teams downloading the wrong data. There were mixed views expressed regarding the use of these environments, as they represent a possible solution to additional challenges regarding data storage, archiving, and destruction, but they also limit the use of the data. Currently there is limited research on the challenges and opportunities of analyzing trial data within TREs.

An additional advantage of using HSD not mentioned during the discussion is the anticipated carbon savings implied by choosing to use HSD. The carbon footprint of clinical trials is a topic of heightened interest, with recent studies showing that participant travel to be a carbon hot-spot [21].

On the other hand, additional disadvantages were not explicitly mentioned during the discussion, including the burden redistribution among the trial team. While gathering data from HSD may reduce the burden imposed on the clinical staff, as it minimizes collection of bespoke data, it may also lead to additional burden for other members of the trial team, for example during the application process and the production of analyzable data sets. Given the participants working experience, some of the issues described can be interpreted as a display of burden redistribution, as effort and complexity is shifted from the clinical setting and the patient cohort to the trialists.

Lastly, the discrepancy between HSD being promoted for its efficiency benefits and the resource-intensive nature of the access process warrants further investigation, including potential cost-effectiveness analyses.

4.2. Strengths and limitations

The strengths of this study include the variety of experiences of the researchers who participated in the focus

groups and interviews. The participants hold different roles within trials and were able to showcase different viewpoints of the process. They were purposefully sampled to have knowledge of accessing different HSD providers from across the UK. In addition, a notable strength of this research is the inclusion of data providers in the participant sample, therefore addressing a gap in the literature by allowing them to share their opinion about the issues identified from a provider side, detail their experience working with the trial teams, and discuss the proposed resolutions.

The small number of participants can be considered a limitation of the current work. Due to the participant's limited availability, there was an increase in the number of one-to-one interviews and a decrease in focus group attendance, therefore limiting the amount of open conversations between participants. While additional challenges and advantages of using HSD may arise within future research, we believe that data saturation was achieved within the scope of this study, with no new themes emerging at the end of the data collection stage, and we consider that no new themes would have arisen from further interviews or focus groups. Being part of a focus group with additional researchers may have led participants to give socially desirable responses. Although efforts were made to prevent it, two participants from the same organization attended the same focus group, which may have influenced the group dynamics. Furthermore, the potential for selection bias can be a limitation as most of the participants were known to the authors to be active within clinical trials and to have experience with HSD. In addition, no detail was captured on the specific role of HSD within participants' trials, such as whether it was used to inform primary or secondary outcomes, which may have provided further context to their experiences.

4.3. Recommendations

Participants suggested improvements both for trial teams and providers that would ease the current issues. The main recommendations, deemed actionable by the researchers, included developing an updated application suitable for RCTs, discussing or involving other researchers with HSD application experience into the trial team early, and building an understanding of the processes between trial teams and HSD providers, by allowing trial teams to learn about the approval processes undergone at the provider's side, while informing the providers of the standards that RCTs have to adhere.

Additional resolutions discussed related to publishing approved HSD applications for trials as examples, making clean datasets available, and discussing with the provider at the design stage of the RCT in order to establish realistic expectations of data and timelines. Such recommendations have been noted previously in literature [13].

Several resources available online were suggested by the participants to help improve the process. These include

written resources [22], working groups dedicated to HSD use in RCTs [23], and online training courses [24]. In addition, where available, reaching out to within institution specialized groups or individuals with knowledge in accessing HSD was recommended.

5. Conclusion

While the use of HSD in RCTs continues to be encouraged, the current access systems are not fit for purpose, resulting in various challenges for trialists. Improvements at HSD provider level and support for trial teams were recommended in order to make HSD access viable for more RCTs. The expected benefits of using HSD, including the improved efficiency and cost decrease, require further research, as the evidence gathered raises questions about the extent to which these advantages may be attained.

This work should inform trial teams of the current obstacles in the process and allow them to better anticipate possible challenges. The recommendations showcase important improvements that may ease the process of accessing HSD, therefore minimizing one of the main problems currently discouraging trial teams from using HSD.

Ethics statement

This study was approved by the Institute of Population Health Research Ethics Committee at the University of Liverpool (Reference: 14,940). Informed written consent was obtained from all participants prior to the interviews or focus groups.

CRediT authorship contribution statement

Alice-Maria Toader: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Carrol L. Gamble:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Susanna Dodd:** Writing – review & editing, Supervision. **Charis Xuan Xie:** Validation, Formal analysis. **Katie Gillies:** Writing – review & editing, Validation, Supervision, Project administration, Methodology.

Consent for publication

This article does not contain any individual person's data as the data has been appropriately anonymized.

Declaration of competing interest

The authors declare that they have no competing interests.

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Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112268>.

Data availability

De-identified data can be made available following a request and further ethical approval.

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